Limits to arbitrage in Indian equities

Stage I: theory, data and the futures-cash band

BS project, Stage I (EC 495) | November 2026

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Guide: Prof. Aditi Chaubal, Department of Economics, IIT Bombay

Agenda

Executive summary

The question and the approach

- Why India, the F&O ban, the model and the frozen test plan

The data

- Bans and the futures-cash basis, 2015 to 2026

The no-arbitrage band

- How wide it is, and what sets it: costs and frictions

Does the ban bind on trading?

- Trading in a ban, the 95% trigger and the 2025 rules

Conclusion and Stage II

Agenda

Executive summary

Executive summary

Futures and cash prices of Indian stocks move in a wide band that costs shape only loosely; the F&O ban binds hard on trading.

A wide, asymmetric band. The basis moves between about -35 and +24 bp around each stock's median. Inside, it drifts; outside, it is pushed back. The lower side reaches further because the correcting trade needs a short sale.

Costs shape it loosely. A new cash-and-carry round trip costs 27 to 31 bp, in the outer tail. The upper tail rose after the 2024 and 2026 tax increases, as predicted, but by more than the cost changes and not unusually against placebo dates.

The ban binds hard on trading. Futures volume falls by about 87% in a ban. Once violations were checked on intraday positions (from March 2020), every futures trade in a ban closed positions.

The 2025 rules cut bans. Against the old rule applied to the same stocks, ban days fell by about three quarters in the first year.

Source: NSE archives, 2010 to 2026; author's analysis

Both primary tests of this stage hold as specified

Pre-specified tests of Stage I; primary tests Holm-corrected with the two of Stage II

Test	Prediction	Result	Verdict
H8 Trading in a ban (primary)	Futures volume falls; trades only close positions	-2.00 log points (s.e. 0.11)	**Supported**
H3 Costs and the band (primary)	Upper edge rises with each tax increase	+6.96 bp (s.e. 0.61); weak against placebo dates	**Supported, weakly**
H1 Frictions and the band	Wider bands where frictions are higher	+3.5 bp per s.d. (p = 0.002), across stocks	**Supported**
Density at the 95% trigger	Not signed by the model	No bunching (p = 0.86)	**Descriptive**

Every detail the plan left open was fixed in a dated log before the test that needed it.

Analyses added after a result (for example, the split by enforcement period) are labelled as such.

Both primary tests of Stage I hold; the cost result is weak once common shocks are allowed for

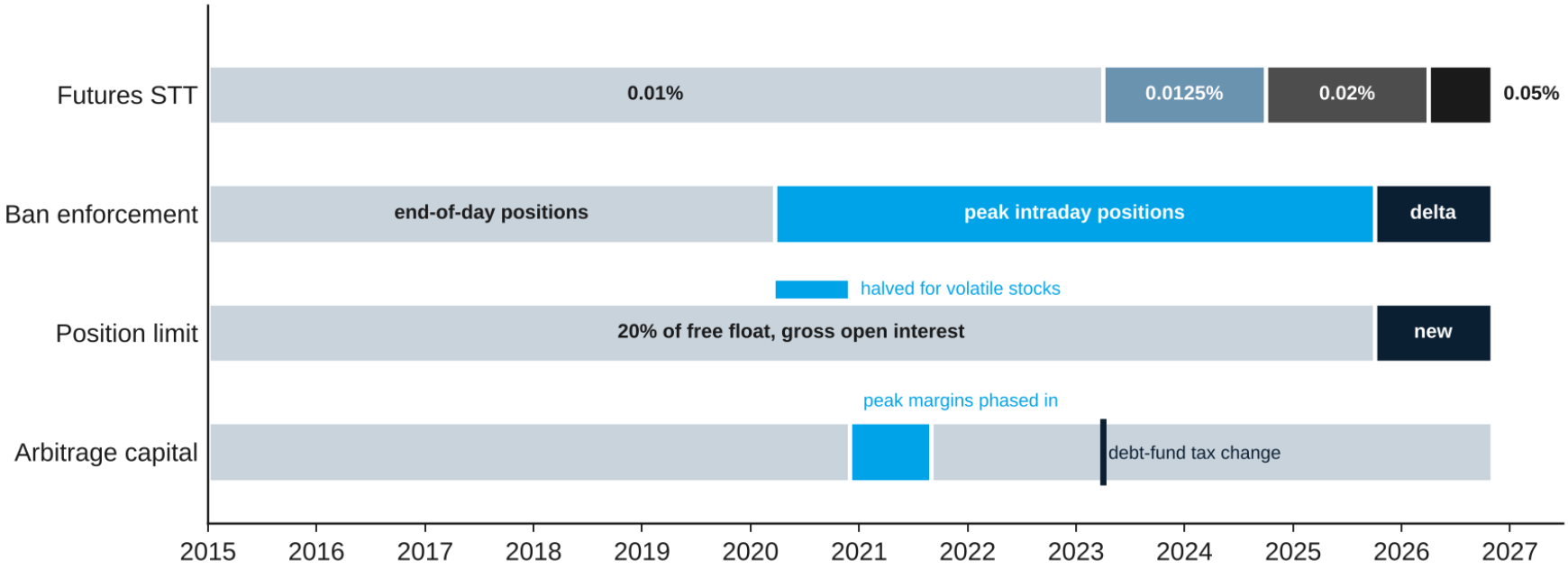
Source: NSE archives; author's analysis

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The question and the approach

India changes almost every limit on a known date

Futures STT, ban enforcement, the position-limit rule and arbitrage capital, 2015 to 2026

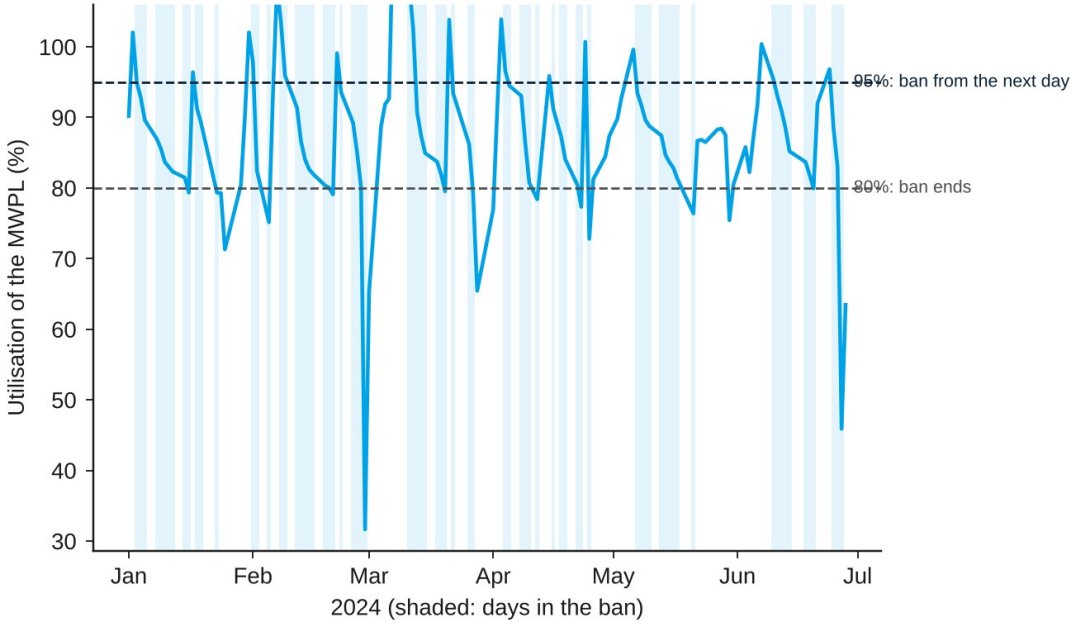


Dated rule changes let me measure each limit instead of assuming it

Source: SEBI circulars and press releases; NSE circulars; AMFI

The F&O ban lets every trader only reduce positions

Utilisation of the position limit for SAIL, the most-banned stock, January to June 2024



How the rule works

Enter the day after open interest exceeds 95% of the market-wide position limit

Leave after it falls to 80%

In between, everyone may only reduce positions, in all expiries

1,788 spells in 141 stocks, 2015 to October 2026

To my knowledge, no study had examined this rule

Source: NSE archives; author's analysis

Stage I builds the foundation; Stage II asks about prices

How strongly do limits to arbitrage bind, and how much mispricing do they explain?

Stage I (this presentation)

A model with every Indian limit as a parameter

A frozen test plan and a data set from public NSE files

How wide is the band within which the futures-cash basis moves?

Do costs and frictions set its edges?

Does the ban bind on trading, as the rule intends?

Stage II

What a ban does to prices: one-sided correction, the level of the basis at ban start

Position limits before a ban: anticipation; the 2025 formula; the 2020 halving

Arbitrage capital: do fund inflows narrow the band?

Short-sale constraints: F&O entry, anomalies, a constraint index for all stocks

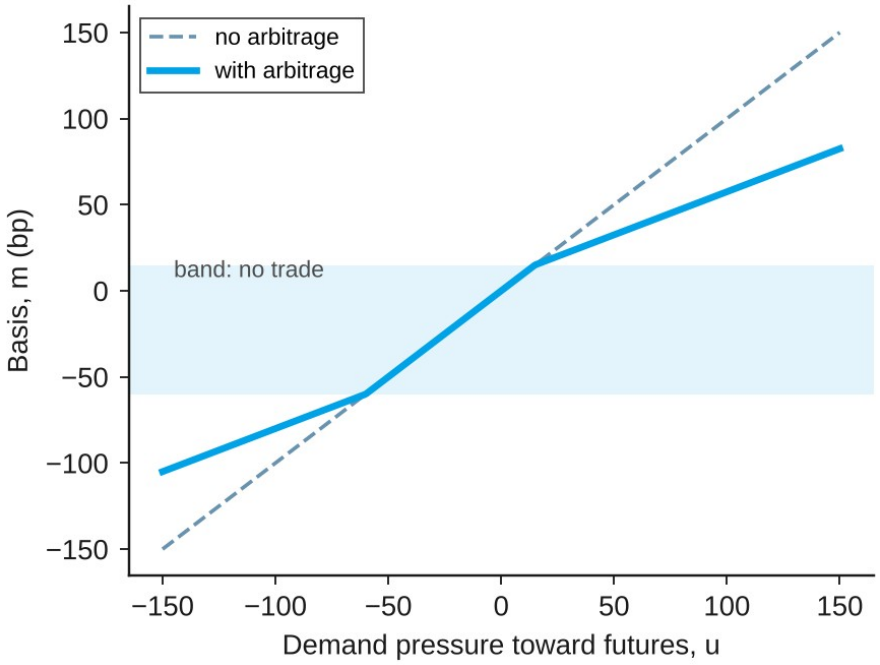
Policy on position limits and their enforcement

Stage I answers three questions: where the band lies, what sets it, and whether bans bind

Source: NSE archives; author's analysis

Model: costs set the band; a ban blocks two things

Equilibrium basis against the demand pressure of outside traders, with illustrative costs



Band	Arbitrage pays only outside **[-c-, c+]** • Costs set the edges; capital and risk set how far the basis travels beyond them
Ban: arbitrage	• Arbitrageurs may only reduce positions • **Rich futures are no longer corrected**
Ban: demand	• New futures demand is blocked; part moves to the stock • **This lowers the basis**

Stage I tests the band and trading in a ban; the price predictions come in Stage II

Source: author's model; illustrative parameters, not estimates

I fixed the tests before looking at outcomes

Five safeguards against fitting theory and data to each other

		What I did	Why
1	**Model**	One model; **every limit a parameter**; each prediction with what would contradict it	Hypotheses are derived, not fitted
2	**Frozen plan**	Tests and signs **fixed with a time stamp** on 5 November 2026	No choosing specifications after seeing results
3	**Data**	Built from **public NSE files** and checked against the exchange's rules	Errors caught before testing
4	**Estimator tests**	Each estimator **recovers known effects** on simulated data	Methods checked before real data
5	**Full reporting**	Primary tests **Holm-corrected**; weak and failed results kept	Nothing is left out

Source: NSE archives; author's analysis

Each prediction states what would contradict it

Predictions of the model and where they are tested

Prediction	Contradicted if	Plan label	Tested in
The basis moves in an asymmetric band, wider below zero	No band; a symmetric band	Band	**Stage I**
Tax increases raise the upper edge, more for larger ones	Edges unrelated to costs	H3 (primary)	**Stage I**
Bands are wider where frictions are higher	No relation	H1	**Stage I**
In a ban futures volume falls and trades only close positions	Futures activity unchanged	H8 (primary)	**Stage I**
In a ban rich futures revert more slowly	No asymmetry, or the opposite	H7 (primary)	Stage II
The level effect follows the composition of trading	Sign unrelated to composition	Level effect	Stage II
Short-sale access speeds price adjustment	No change, or the opposite	H6 (primary)	Stage II

Source: pre-analysis plan, frozen 5 November 2026

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The data

The data come entirely from public NSE files

Sources, coverage and the main checks

Bans and limits	• Daily open interest, position limits and ban lists, February 2010 to October 2026 • Next-day flags match the official ban list for 11,753 of 11,764 flags
Prices and basis	• Futures and cash bhavcopies 2015 to 2026; basis from volume-weighted prices • Settlement prices avoided: they are set at cost of carry on thin days
Corporate events	• Dividends subtracted at present value • Contracts with a dividend or other event in their life excluded
Lending, rates	• Daily stock-lending trades, 2015 to 2026 • Three-month interbank rate for the cost of carry

1,788
ban spells

9,416
ban stock-days

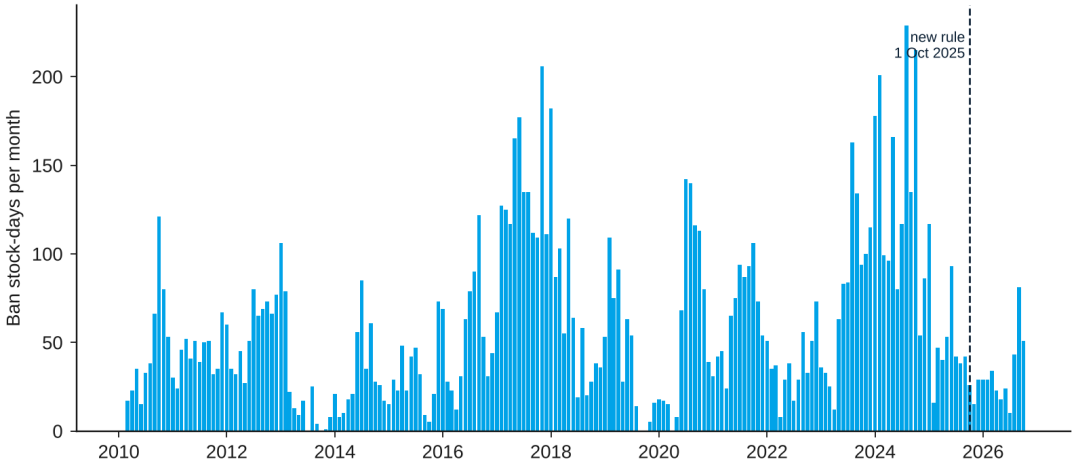
486,040
stock-days of basis

371
stocks with futures

Source: NSE archives; OECD via FRED; author's analysis

Bans come in waves and fell under the new rule

Stock-days in the ban per month, February 2010 to October 2026



What the chart shows

1,788 spells in 141 stocks, 2015 to October 2026

Typical spell: 4 trading days before the reform, 9 after it

1 October 2025: every stock's limit cut, to a median of 73% of the old value

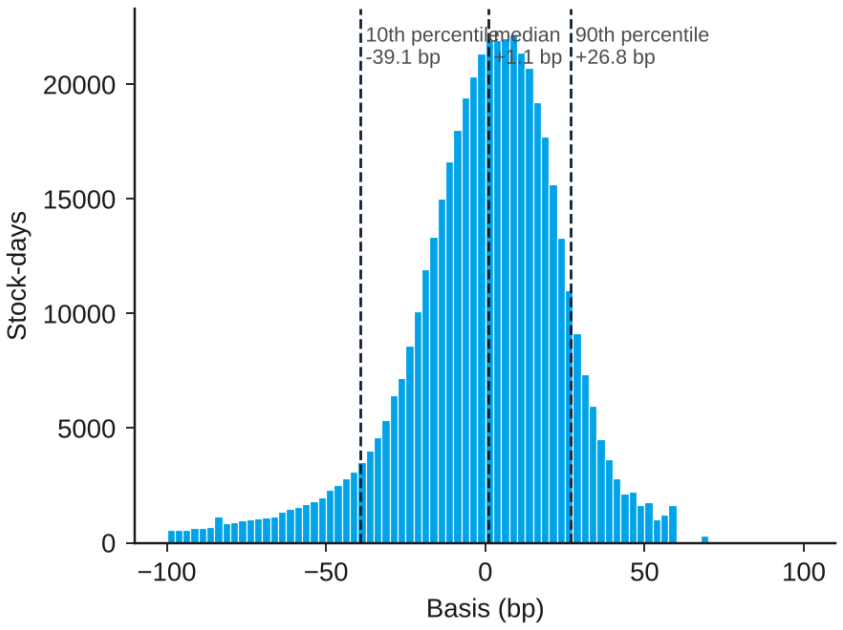
Bans are frequent enough to study, and concentrated in a few stocks

Source: NSE securities-in-ban lists; author's analysis

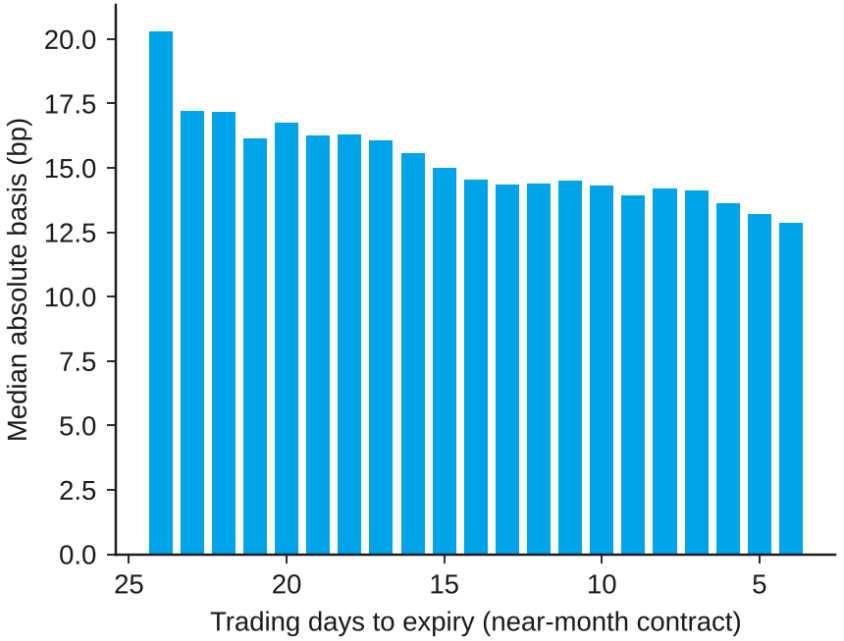
The basis is centred near zero with a long lower tail

Futures-cash basis of the analysis sample, 2015 to 2026

Median +1.1 bp; cheap futures go further from zero



The basis narrows as expiry approaches



Cheap futures are further from zero, as expected when the correcting trade needs a short sale

Source: NSE archives; author's analysis

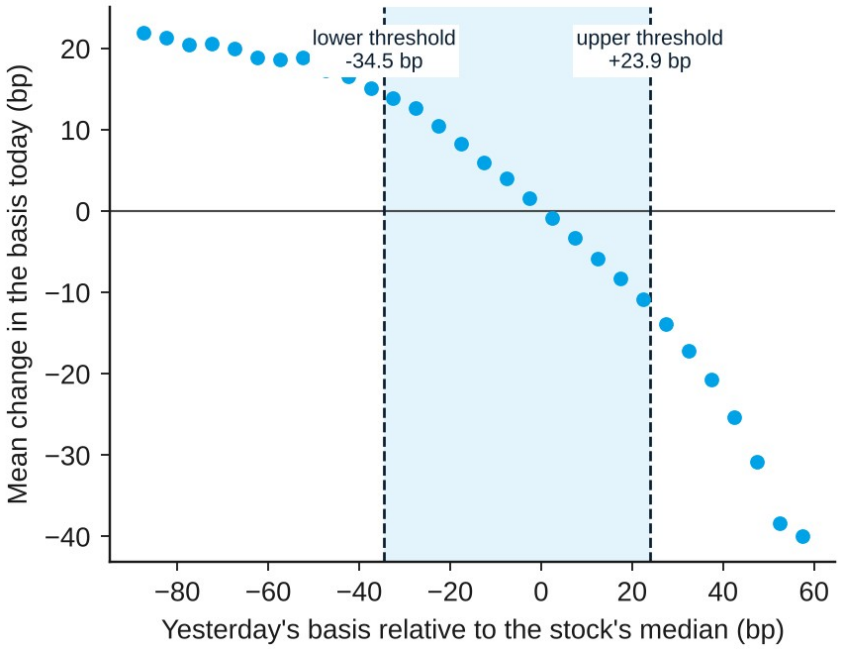
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The no-arbitrage band

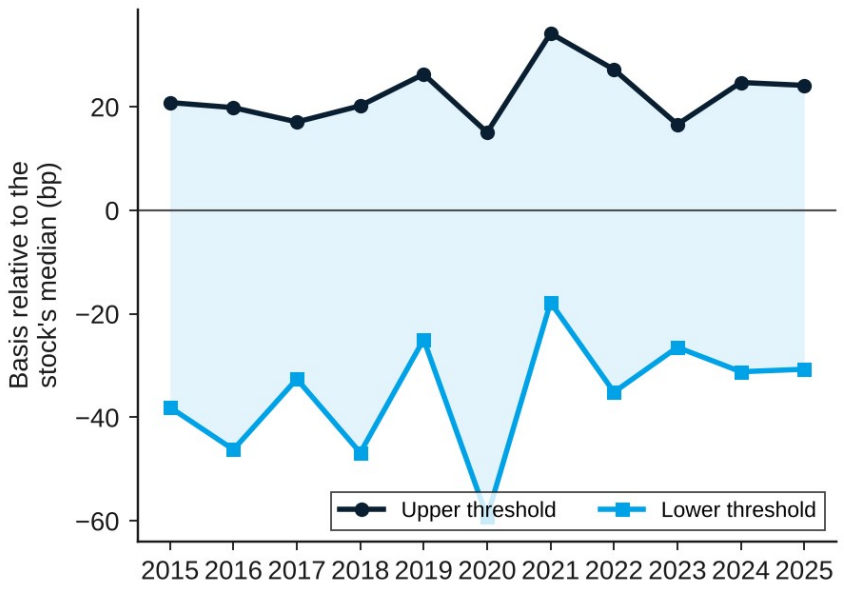
The basis moves in a wide, asymmetric band

Pooled threshold model, non-ban days, 2015 to 2026, 360 stocks

Pull back grows with the deviation; band -34.5 to +23.9 bp



Stable across years, apart from March 2020



The band is about 58 bp wide, far wider than intraday bands found in earlier Indian studies

Source: NSE archives; author's analysis

The band's location is robust; its speed is not

Reversion slopes by regime, before and after a correction for noise

	Lower regime	Middle	Upper regime
Threshold (bp)	below -34.5		above +23.9
Share of days	12%	76%	12%
Slope, threshold model	-0.08	-0.47	-0.81
Slope, noise-corrected	-0.12	-0.19	-0.14

What this means

Daily volume-weighted prices contain noise that looks like fast reversion

Corrected, the three slopes are small and similar

Smooth-transition models fit almost as well; a symmetric one fits worse

Read the thresholds, not the speeds, as the main evidence on the band

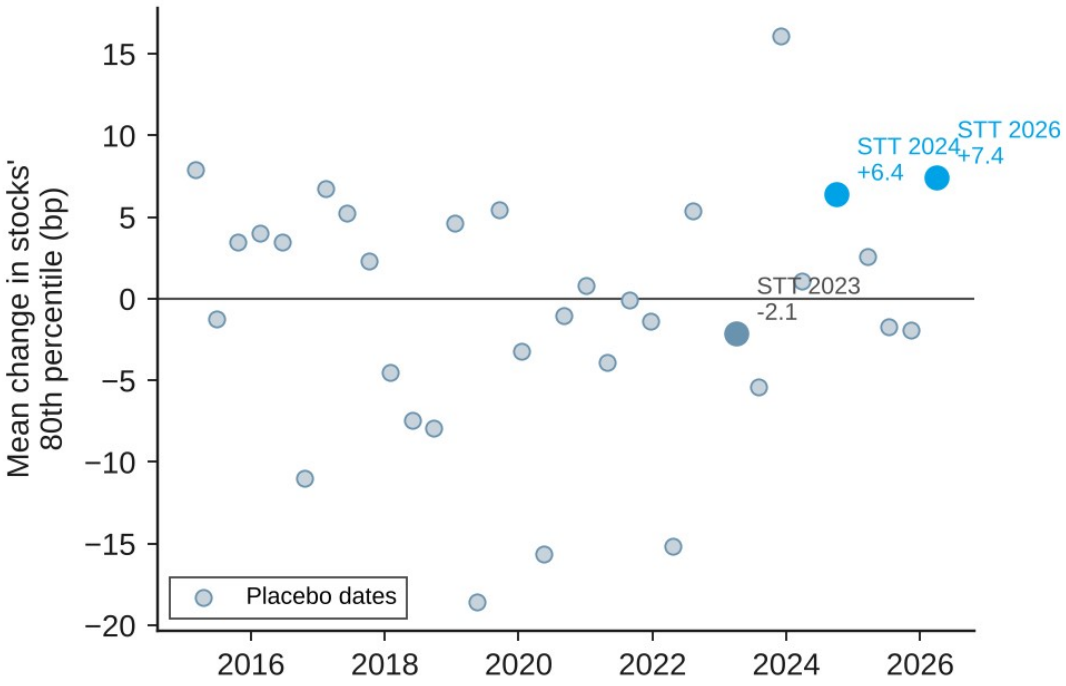
Source: NSE archives; author's analysis

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What sets the band

Tax rises lifted the upper tail, but not unusually

Change in stocks' 80th percentile at the STT dates and at 30 placebo dates (bp)



What the test shows

Predicted: the upper edge rises with each tax increase, more for larger ones

Found: +6.4 bp (2024) and +7.4 bp (2026); ordered as the cost changes

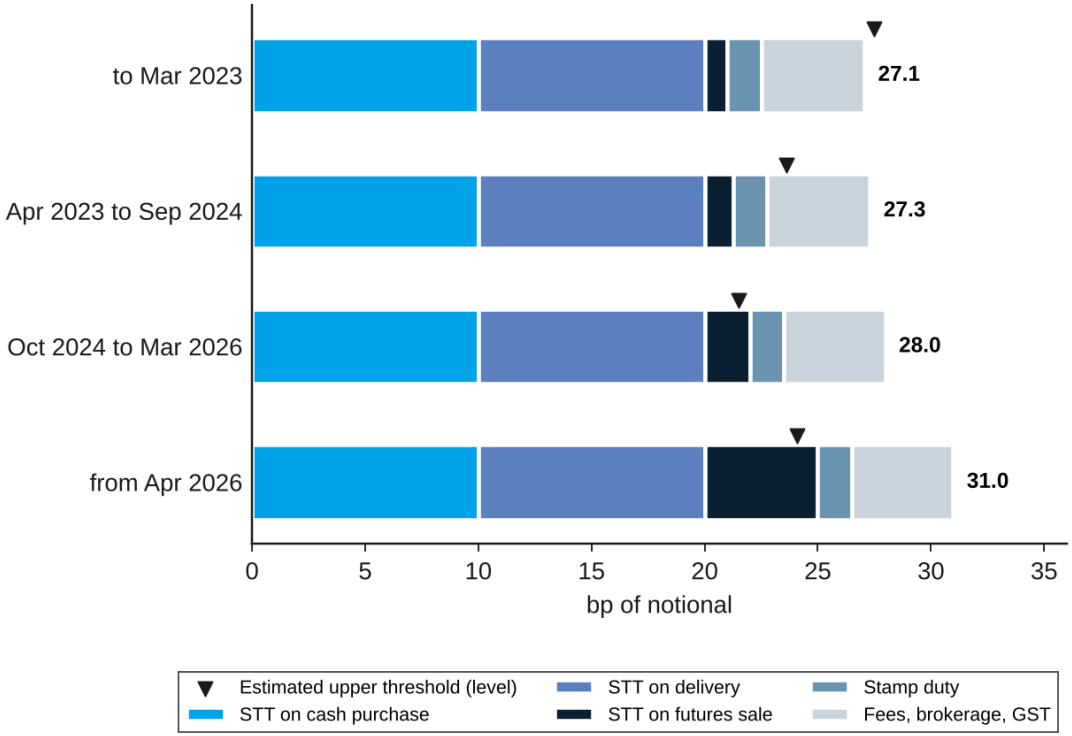
But: larger than the cost changes, the median moved too, and placebo dates move as much

Costs push the band the predicted way but do not set its edges tightly

Source: NSE archives; author's analysis

A new position costs more than the estimated edge

Round trip of an overnight cash-and-carry position, by STT period (bp)



What the chart shows

New position: 27 to 31 bp, between the 90th and 95th percentiles of the basis

Estimated upper edge: +22 to +28 bp

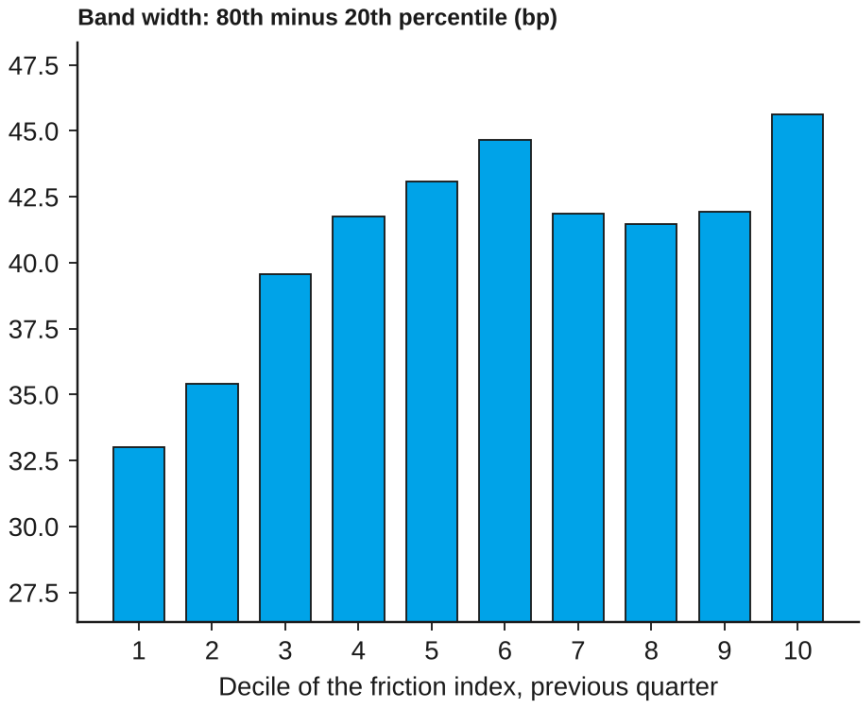
Rolling an existing position: only 4 to 8 bp

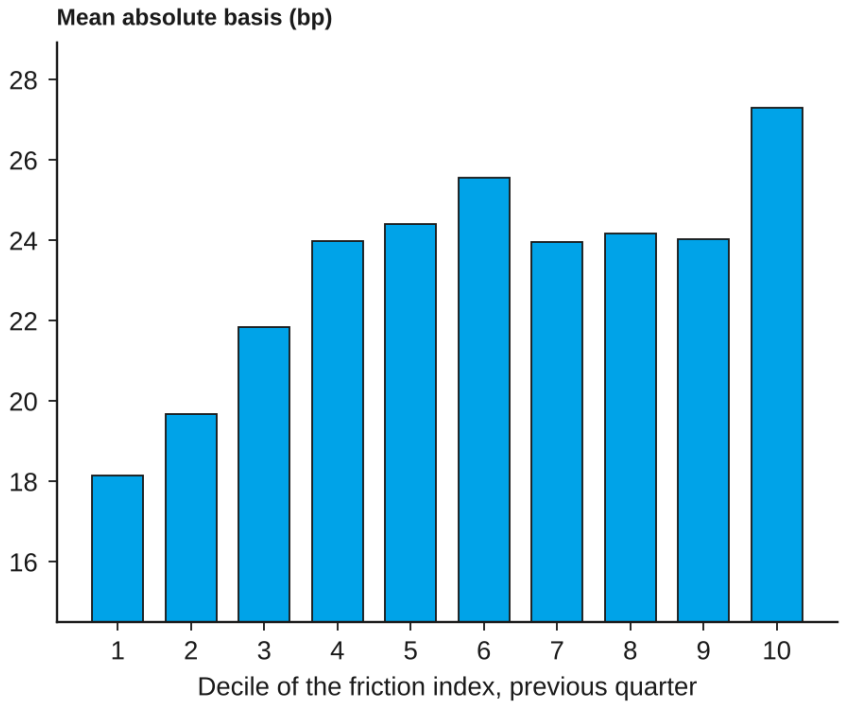
New capital enters only at large gaps; funds that roll lean against small ones

Source: NSE circulars on STT and transaction charges; author's analysis

Bands are wider in stocks that are harder to trade

Band width and mean absolute basis by decile of a friction index (previous quarter)





The link holds across stocks, not within a stock over time

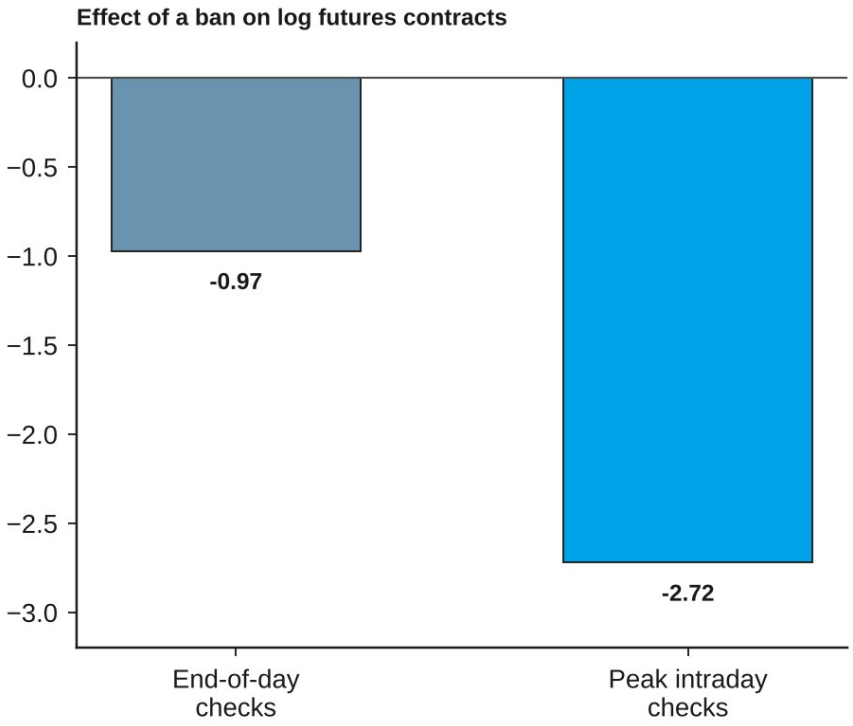
Source: NSE archives; author's analysis

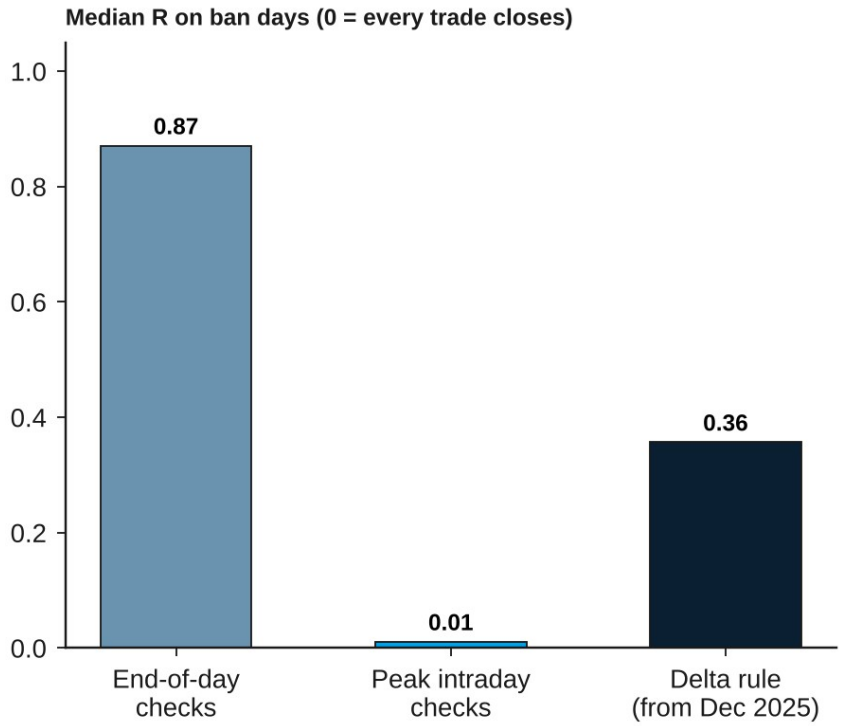
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Does the ban bind on trading?

In a ban, futures trading collapses

Effect on log futures contracts, and median R on ban days, by enforcement regime



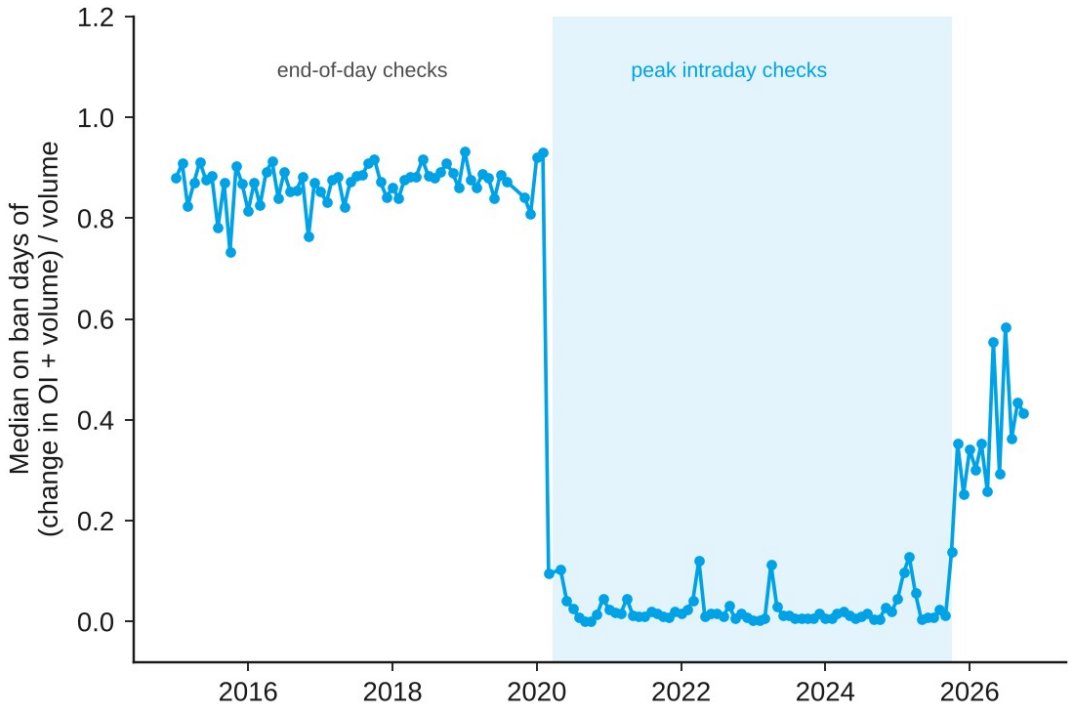


Futures volume falls by about 87% in a ban; cash trading rises and turns toward delivery

Source: NSE archives; author's analysis

Intraday checks made the rule bind trade by trade

Monthly median on ban days of R = (change in open interest + volume) / volume



What the chart shows

To 22 March 2020 violations were judged on end-of-day positions: median R 0.87

From 23 March 2020 on peak intraday positions: median R 0.011

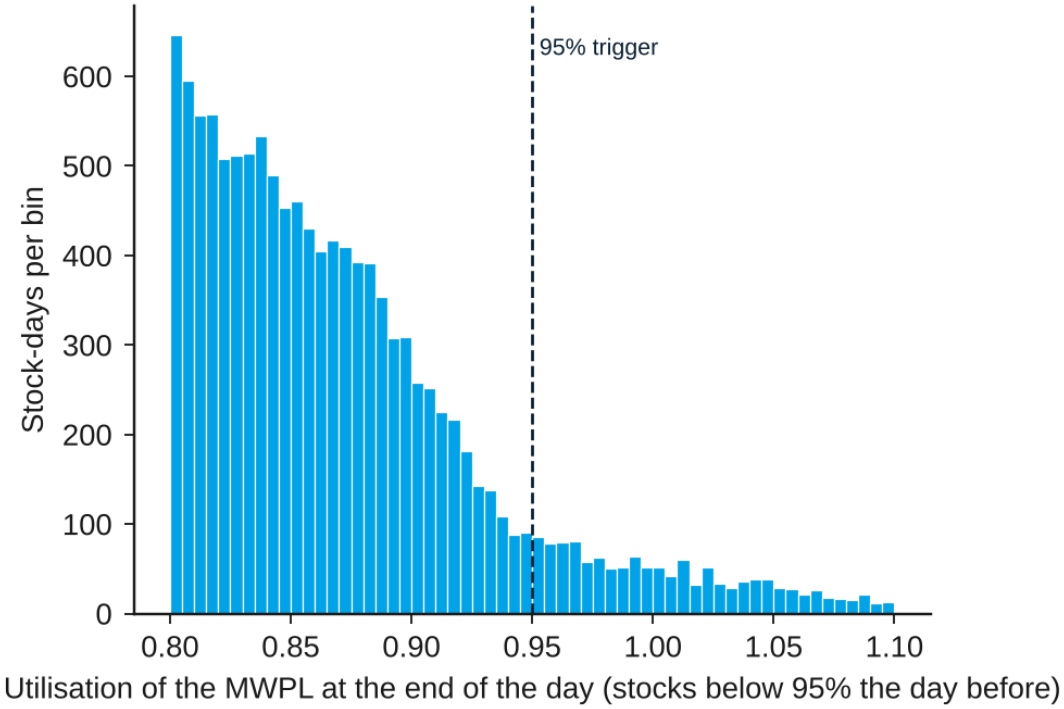
From October 2025 the delta rule allows offsetting trades: 0.19 to 0.36

The rule changes trading as designed only when positions are checked during the day

Source: NSE archives; author's analysis

Traders do not hold back below the 95% trigger

End-of-day utilisation of stocks that were below 95% the day before, 2015 to September 2025



What the chart shows

No bunching just below 95% (p = 0.86)

Utilisation speeds up near the trigger rather than stalling

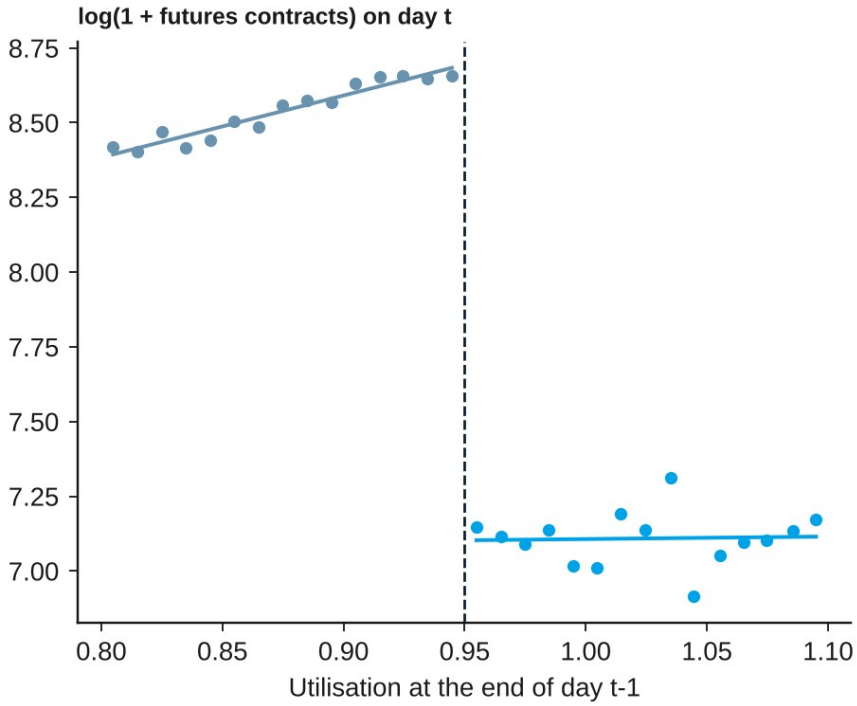
Chance of crossing next day: 7% at 85% utilisation, about half at 93 to 95%

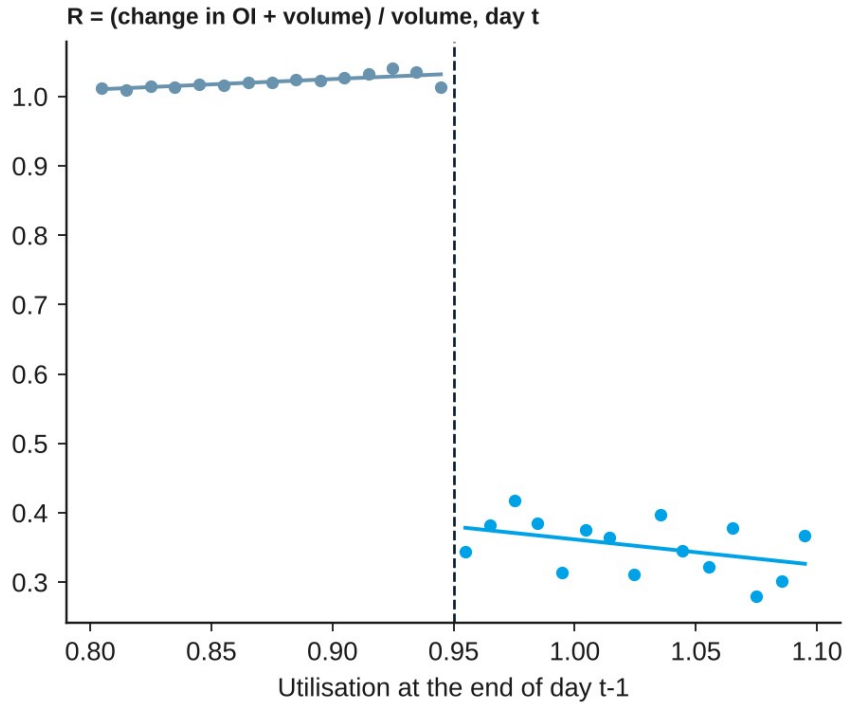
There is no sign that traders hold back to avoid a ban

Source: NSE archives; author's analysis

Trading breaks sharply at the 95% trigger

Binned means by utilisation at the end of the previous day, stocks not in a ban the day before



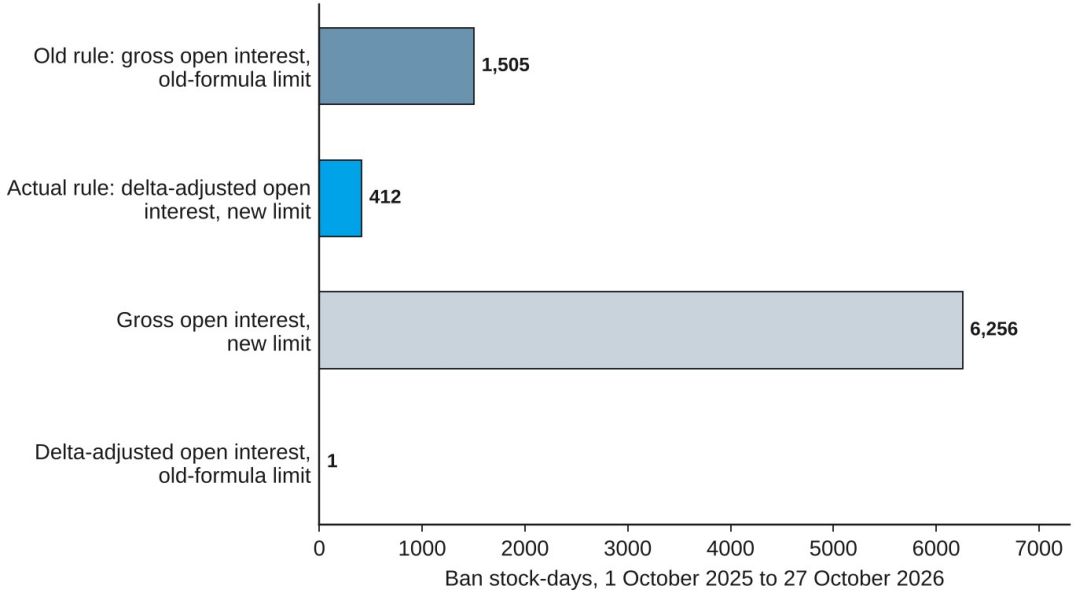


Crossing 95% puts a stock in the ban the next day in 99.7% of cases

Source: NSE archives; author's analysis

The 2025 rules cut ban days by about three quarters

Ban stock-days, 1 October 2025 to 27 October 2026, actual and counterfactual rules



What the reform did

Aim: SEBI set out to reduce 'spurious' bans

Before 2025 bans came from rising open interest; only 0.4% from a falling limit

Delta-adjusted open interest removes almost all bans; the lower limit alone would have quadrupled them

Net: 412 ban days against 1,505 under the old rule

The reform met its stated aim on the number of bans

Source: NSE archives; author's analysis

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Conclusion and Stage II

What Stage I shows

The three questions of this stage and their answers

Question	Answer	Main evidence
How wide is the band?	Wide and asymmetric	-35 to +24 bp around each stock's median; linearity rejected; stable over time
Do costs set its edges?	Loosely	Upper tail up after 2024 and 2026 tax rises, but not unusually; new position costs 27 to 31 bp
Do frictions widen it?	Across stocks	+3.5 bp per s.d. of a friction index; no effect within a stock
Does the ban bind on trading?	Yes, strongly	Futures volume -87%; every trade closes positions under intraday checks; 2025 rules cut bans by three quarters

Limits bind hard on trading; what they do to prices is the question of Stage II

Source: NSE archives; author's analysis

Four limitations of this stage

Limitation	What could go wrong	...and what I did
Daily data	Intraday arbitrage is out of scope, and noise inflates measured reversion	Noise-corrected estimates reported alongside
Positions unobserved	The arbitrage book in the model is not seen in public data	Edges tested through changes, not levels
Few cost events	Three tax increases, one on the same day as a fund tax change	Placebo dates calibrate the test
Short post-reform period	Only 44 ban spells since October 2025	Reform effect on prices left to Stage II

Intraday data and a longer post-reform sample would sharpen the tests

Source: NSE archives; author's analysis

Stage II asks what the limits do to prices

The plan for the second stage

1	**The ban and prices**	One-sided correction in a ban (H7); the basis when a ban starts; placebo bans
2	**Limits before a ban**	Is the basis richer as utilisation nears 95%? The 2025 formula; the 2020 halving
3	**Arbitrage capital**	Do inflows into arbitrage funds narrow the band?
4	**Short-sale access**	F&O entry and exit (H6); anomalies; a constraint index for all stocks
5	**Policy**	Design and enforcement of position limits

Stage I's model, plan and data carry over unchanged

Source: NSE archives; author's analysis

Limits to arbitrage in Indian equities

Thank you

BS project, Stage I (EC 495) | November 2026

Prince Kumar (23B3032)

Guide: Prof. Aditi Chaubal, Department of Economics, IIT Bombay

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Appendix

The model's propositions behind each test

Proposition	Statement	Tested by
1 Band	Arbitrage pays only outside [-c-, c+]; inside, the basis follows demand	Band estimation
2 What moves it	Costs set the edges; capital and risk set how far the basis goes beyond	H3 (Stage I); H4 (Stage II)
3 Existing book	The no-trade region moves with the existing position	Design of the cost test
4 One-sided	In a ban rich futures are not corrected; cheap ones are while positions last	H7 (Stage II)
5 Level effect	The ban's effect on the level has no fixed sign	Stage II
6 Trading	If the rule binds on every trade, each trade closes positions	H8 (Stage I)

Source: author's model; proofs in the Stage I report

How the data were built and checked

Bans	• Next-day flags agree with the official list for 11,753 of 11,764 flags • 95% entry and 80% exit rules hold for 99.4% and 99.7% of spells
Basis	• Volume-weighted futures and cash prices, net of dividends and carry • Futures price inside the day's range for 99.8% of contracts
Sample	• Near contract, next one in the last three trading days; at least five contracts • First 20 days after F&O entry excluded; winsorised at 1% and 99% within each year
Reproduction	• One script sequence produces every estimate; a full rerun reproduced every number • Each estimator first recovered known effects on simulated data

Source: author's analysis; Stage I report, Section 5 and Appendix B